

MATH 280 MATLAB Project #3 - Multivariable Function and Optimization Analyzer



When the project is completed, rename the end of the file to (Completed Implementation).

$$f(x,y) = 2x^2 + 3xy + 6x + 7y$$

$$\frac{\partial f}{\partial x} = 4x + 3y + 6$$

$$\frac{\partial^2 f}{\partial x^2} = 4$$

$$\frac{\partial f}{\partial y} = 3x + 7$$

$$\frac{\partial^2 f}{\partial y^2} = 0$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = 3$$

PROJECT OUTLINE

Core Objective

Develop a MATLAB application that allows users to visualize and analyze functions of two and three variables. The application provides interactive tools for computing partial derivatives, directional derivatives, gradient vectors, tangent planes, extrema, saddle points, and constrained optimization using Lagrange Multipliers through an intuitive graphical user interface.

Rather than acting as a symbolic calculator, the application serves as an interactive Multivariable Calculus analysis and optimization tool.

Concepts Covered

Chapter 14 Partial Derivatives

- 14.1 Functions of Several Variables
- 14.2 Limits and Continuity in Higher Dimensions
- 14.3 Partial Derivatives
- 14.4 Chain Rule
- 14.5 Directional Derivatives and Gradient Vectors
- 14.6 Tangent Planes and Differentials
- 14.7 Extreme Values and Saddle Points
- 14.8 Lagrange Multipliers

Learning Objectives

By completing this project you will practice:

1. multivariable functions
2. surface visualization
3. numerical differentiation
4. symbolic differentiation
5. the concept of optimization from an engineering perspective
6. engineering visualization

7. interactive mathematical modeling
8. MATLAB App Designer

Required Features

1. Multivariable Function Explorer in MATLAB

User enters a function of two or three variables and MATLAB displays:

- 3D surface
- contour map
- color map
- wireframe view

2. Limit and Continuity Explorer in MATLAB

User selects the approach point value,

And MATLAB displays:

- approaching paths
- function values
- continuity result

This feature allows users to compare multiple paths approaching the same point to visualize when limits do or do not exist.

3. Partial Derivative Calculator in MATLAB

Given a function of two or three variables and the variable to differentiate wrt to, MATLAB computes

- First partial derivatives
- Second partial derivatives
- Mixed partial derivatives (e.g. f_{xy})

And displays:

- symbolic result
- numerical evaluation
- derivative surface

4. Chain Rule Analyzer in MATLAB

User enters outer and inner functions,
and MATLAB computes:

- composed derivatives
- intermediate calculations
- final derivative

MATLAB must display the computation step-by-step.

5. Gradient and Directional Derivative Explorer in MATLAB

The user enters evaluation point and direction vector and

MATLAB computes:

- gradient vector
- unit direction vector
- directional derivative

And displays:

- surface
- gradient arrow
- direction arrow
- steepest ascent path

6. Tangent Plane Generator in MATLAB

The user selects a point on a function plane.

MATLAB computes:

- tangent plane equation
- normal vector
- differential approximation

And displays:

- surface
- tangent plane
- point of tangency

7. Optimization Analyzer in MATLAB

MATLAB finds/calculates:

- local maxima
- local minima
- saddle points

And displays:

- classified critical points
- Hessian information
- highlighted locations on the surface

8. Lagrange Multiplier Solver in MATLAB

User enters the function and constraint

MATLAB calculates:

- Lagrange multiplier
- optimal point(s)
- constrained maximum
- constrained minimum

And displays:

- constraint curve
- surface
- optimal solution

9. Engineering Optimization Demonstrations in MATLAB

Include built-in examples such as

- minimizing material usage
- maximizing volume
- optimizing manufacturing dimensions
- heat distribution surfaces
- cost optimization

This is almost like calculus with economic applications.

Users can switch between these examples from a drop-down menu or other menu and compare them with their own custom functions.

Necessary MATLAB Files For This Project/File Structure

MVC_Project3/
main.mlapp
main.m

surfaceGenerator.m
limitAnalyzer.m
partialDerivativeCalculator.m
chainRuleCalculator.m
gradientCalculator.m
directionalDerivativeCalculator.m
tangentPlaneGenerator.m
optimizationAnalyzer.m
lagrangeMultiplierSolver.m
plotUtilities.m
README.md

Useful MATLAB Commands For This Project

fsurf()
fcontour()
mesh()
surf()
meshgrid()
gradient()
diff()
jacobian() - matrix of first partial derivatives
hessian() - matrix of second partial derivatives
solve()
fminsearch()
contour3()
quiver3()
scatter3()
hold on
axis equal

`view()`
`uifigure()`
`uiaxes()`
`uibutton()`
`uidropdown()`
`uislider()`
`uitable()`
`uilabel()`

Additional MATLAB Pseudocode

Surface Generation

Receive function
Generate meshgrid
Evaluate function
Plot surface

Partial Derivatives

Receive function
Differentiate
Evaluate at point
Display results

Gradient Calculation

Receive function
Compute partial derivatives
Construct gradient vector
Display vector on surface

Tangent Plane

Receive point

Compute gradient
Generate tangent plane
Overlay with surface

Optimization

Receive function
Find critical points
Classify each point
Display maxima
Display minima
Display saddle points

Lagrange Multipliers

Receive objective function
Receive constraint
Solve system
Display optimal points
Visualize solution

GUI FUNCTIONALITY

1. Function input field supporting custom multivariable expressions
2. Sliders for selecting evaluation points on the surface
3. Toggle buttons to display:
4. Gradient vectors
5. Tangent planes
6. Contour lines
7. Critical points
8. Constraint curves

Drop-down menu for predefined engineering examples

9. Tabs separating:

10. Visualization
11. Differentiation
12. Optimization
13. Constrained Optimization
14. Interactive rotation and zoom controls
15. Export plots and numerical results

MAIN MENU

Multivariable Function Analyzer

1. Plot Function
2. Analyze Limits
3. Partial Derivatives
4. Chain Rule
5. Gradient Vector
6. Tangent Plane
7. Optimization
8. Lagrange Multipliers
9. Export Results
10. Exit

GUI LAYOUT

Primary Interface

Left Panel

Input fields

- Function $f(x,y)$ (or $f(x,y,z)$)
- Variable ranges
- Evaluation point
- Constraint equation
- Direction vector

Buttons

- Plot
- Analyze
- Reset

Center

Interactive 3D surface plot

Right Panel

Displays

- Partial derivatives
- Gradient vector
- Directional derivative
- Tangent plane equation
- Critical points
- Function value

Bottom

Buttons

- Rotate
- Zoom
- Export
- Animate
- Save Figure

FINAL DELIVERABLES

- MATLAB App Designer project (.mlapp) providing the primary graphical interface.
- MATLAB source files organized into reusable modules for visualization, differentiation, and optimization.
- A README describing the mathematical concepts, application architecture, GUI features, and instructions for running the software.
- Sample demonstrations illustrating surface visualization, limit analysis, partial derivatives, gradient vectors, tangent planes, optimization, saddle point classification, and constrained optimization with Lagrange multipliers.

multivariable calculus

$$\iiint f(x,y,z) \frac{\partial f}{\partial y} \frac{\partial f}{\partial x} 7i + 4j + 3z$$